```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
import scipy.stats as stats
import scipy.stats as ttest_ind
warnings.filterwarnings('ignore')
```

```
# Correct direct download link
file_path = 'https://drive.google.com/uc?id=1IjEP6XqVFNZfYvFGDyC0SHWmwr1nsA7a'

# Load the CSV file
df = pd.read_csv(file_path)

# Display the first few rows
df.head()
```

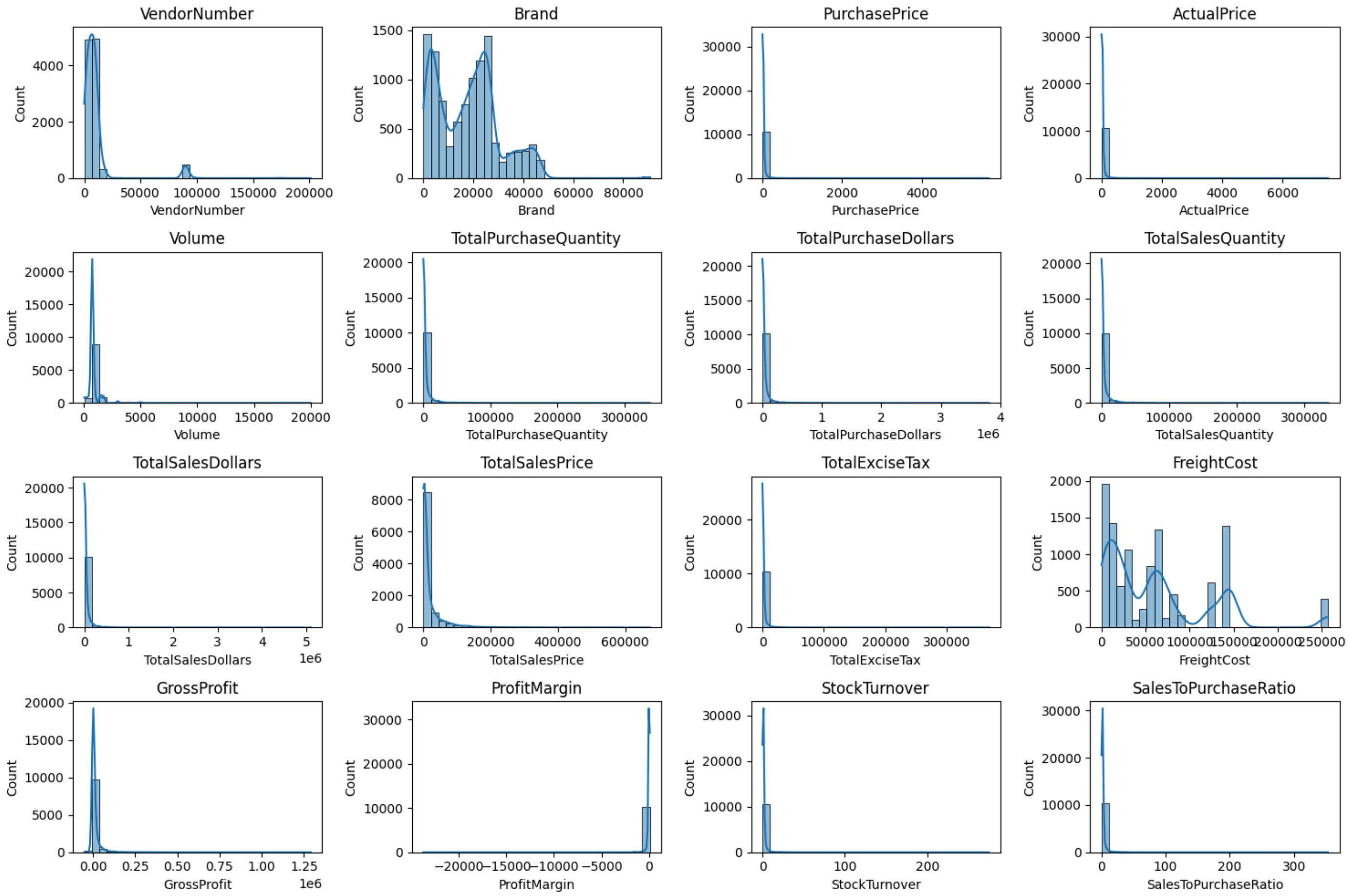
	VendorNumber	VendorName	Brand	Description	PurchasePrice	ActualPrice	Volume	TotalPurchaseQuantity	TotalPurchaseDollars	Tota
0	1128	BROWN-FORMAN CORP	1233	Jack Daniels No 7 Black	26.27	36.99	1750.0	145080	3811251.60	
1	4425	MARTIGNETTI COMPANIES	3405	Tito's Handmade Vodka	23.19	28.99	1750.0	164038	3804041.22	
2	17035	PERNOD RICARD USA	8068	Absolut 80 Proof	18.24	24.99	1750.0	187407	3418303.68	
3	3960	DIAGEO NORTH AMERICA INC	4261	Capt Morgan Spiced Rum	16.17	22.99	1750.0	201682	3261197.94	
4	3960	DIAGEO NORTH AMERICA INC	3545	Ketel One Vodka	21.89	29.99	1750.0	138109	3023206.01	

```
# Basis EDA(Exploratory Data Analysis)
# Summary Statistics
df.describe().T
```

	count	mean	std	min	25%	50%	75%	max
VendorNumber	10692.0	1.065065e+04	18753.519148	2.00	3951.000000	7153.000000	9552.000000	2.013590e+05
Brand	10692.0	1.803923e+04	12662.187074	58.00	5793.500000	18761.500000	25514.250000	9.063100e+04
PurchasePrice	10692.0	2.438530e+01	109.269375	0.36	6.840000	10.455000	19.482500	5.681810e+03
ActualPrice	10692.0	3.564367e+01	148.246016	0.49	10.990000	15.990000	28.990000	7.499990e+03
Volume	10692.0	8.473605e+02	664.309212	50.00	750.000000	750.000000	750.000000	2.000000e+04
TotalPurchaseQuantity	10692.0	3.140887e+03	11095.086769	1.00	36.000000	262.000000	1975.750000	3.376600e+05
TotalPurchaseDollars	10692.0	3.010669e+04	123067.799627	0.71	453.457500	3655.465000	20738.245000	3.811252e+06
TotalSalesQuantity	10692.0	3.077482e+03	10952.851391	0.00	33.000000	261.000000	1929.250000	3.349390e+05
TotalSalesDollars	10692.0	4.223907e+04	167655.265984	0.00	729.220000	5298.045000	28396.915000	5.101920e+06
TotalSalesPrice	10692.0	1.879378e+04	44952.773386	0.00	289.710000	2857.800000	16059.562500	6.728193e+05
TotalExciseTax	10692.0	1.774226e+03	10975.582240	0.00	4.800000	46.570000	418.650000	3.682428e+05
FreightCost	10692.0	6.143376e+04	60938.458032	0.09	14069.870000	50293.620000	79528.990000	2.570321e+05
GrossProfit	10692.0	1.213238e+04	46224.337964	-52002.78	52.920000	1399.640000	8660.200000	1.290668e+06
ProfitMargin	10692.0	-inf	NaN	-inf	13.324515	30.405457	39.956135	9.971666e+01
StockTurnover	10692.0	1.706793e+00	6.020460	0.00	0.807229	0.981529	1.039342	2.745000e+02
SalesToPurchaseRatio	10692.0	2.504390e+00	8.459067	0.00	1.153729	1.436894	1.665449	3.529286e+02

```
# Distribution Plots for Numerical Columns
num_colms = df.select_dtypes(include=np.number).columns
```

```
plt.figure(figsize=(15,10))
for i, col in enumerate(num_colms):
  plt.subplot(4,4,i+1)
  sns.histplot(df[col],kde=True, bins=30)
  plt.title(col)
plt.tight_layout()
plt.show()
```




```
# Filter the data
df = df[
    (df['GrossProfit'] > 0) &
    (df['ProfitMargin'] > 0) &
    (df['TotalSalesQuantity'] > 0)
]

# Show the cleaned data
df.head()
```

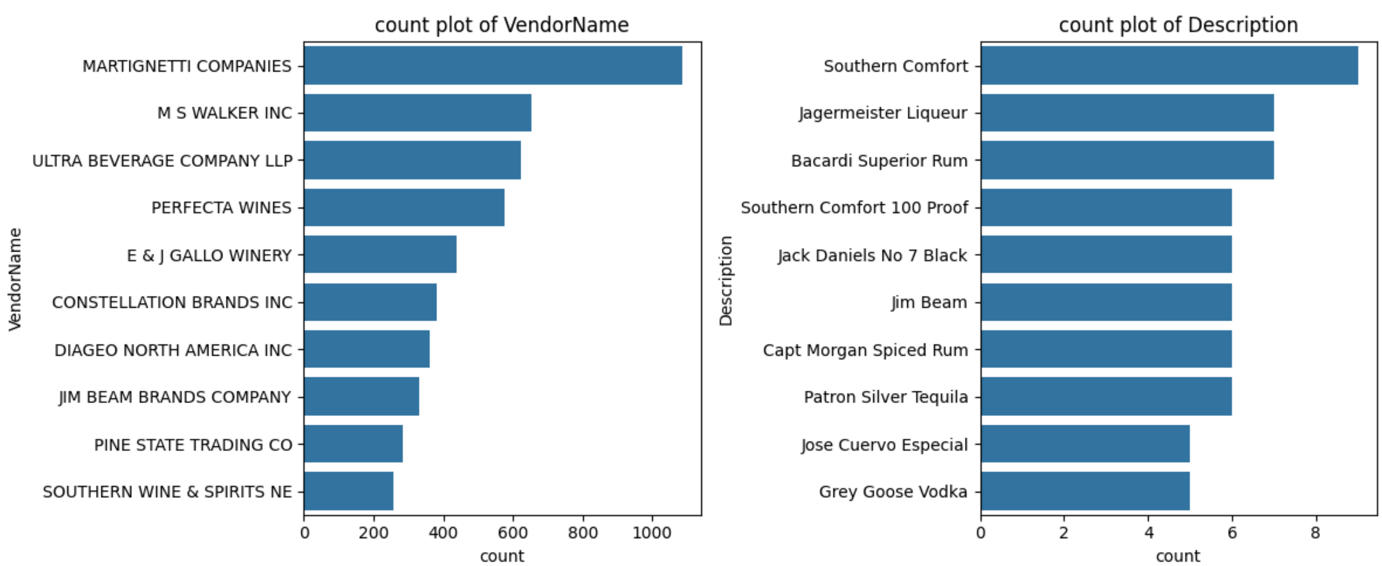
	VendorNumber	VendorName	Brand	Description	PurchasePrice	ActualPrice	Volume	TotalPurchaseQuantity	TotalPurchaseDollars	Tota
0	1128	BROWN-FORMAN CORP	1233	Jack Daniels No 7 Black	26.27	36.99	1750.0	145080	3811251.60	
1	4425	MARTIGNETTI COMPANIES	3405	Tito's Handmade Vodka	23.19	28.99	1750.0	164038	3804041.22	
2	17035	PERNOD RICARD USA	8068	Absolut 80 Proof	18.24	24.99	1750.0	187407	3418303.68	
3	3960	DIAGEO NORTH AMERICA INC	4261	Capt Morgan Spiced Rum	16.17	22.99	1750.0	201682	3261197.94	
4	3960	DIAGEO NORTH AMERICA INC	3545	Ketel One Vodka	21.89	29.99	1750.0	138109	3023206.01	

Next steps: Generate code with df | View recommended plots | New interactive sheet

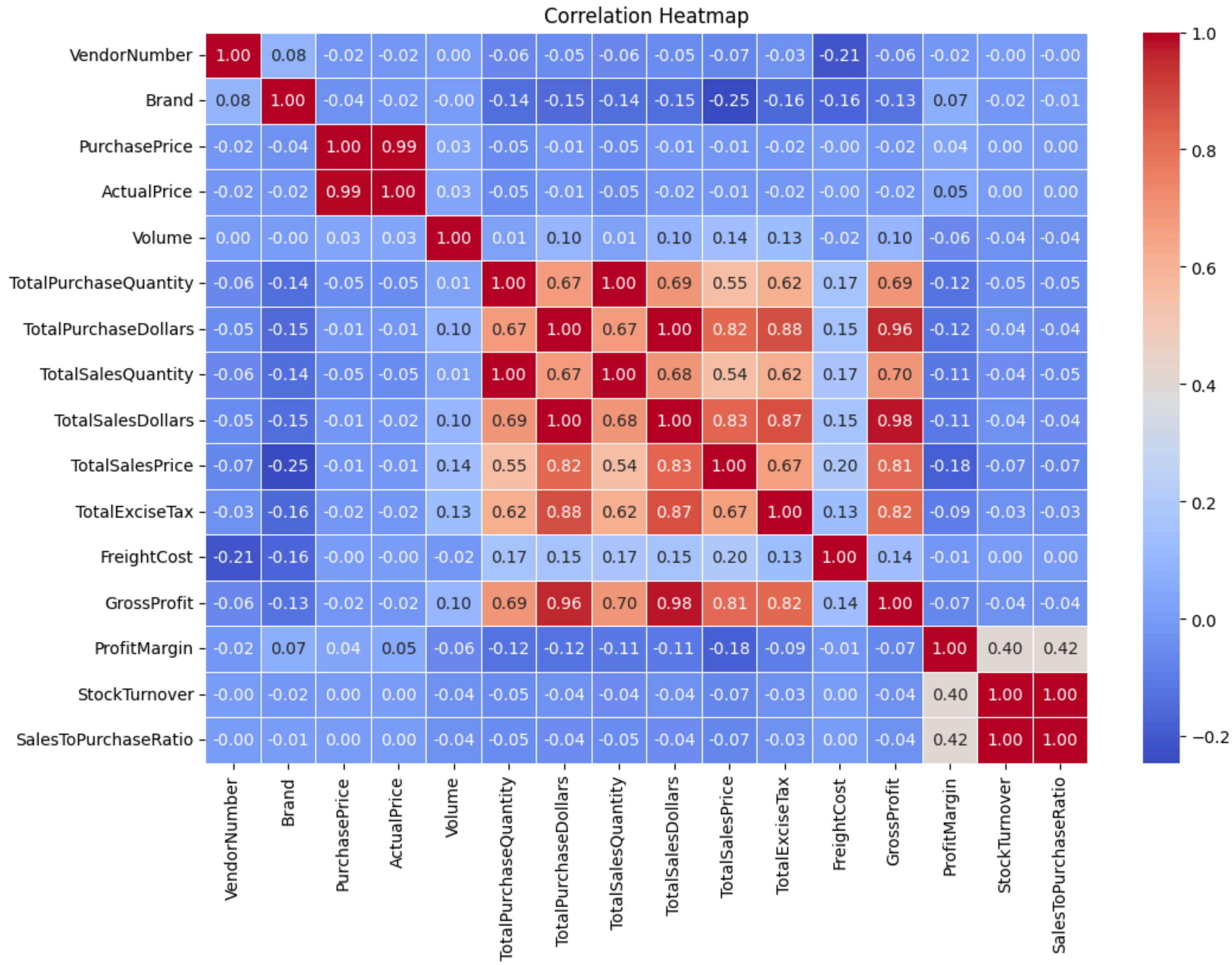
```
df.shape
```

```
(8565, 18)
```

```
# count plot for categorical columns
cat_colms = df[['VendorName','Description']]

plt.figure(figsize=(12, 5))
for i, col in enumerate(cat_colms):
  plt.subplot(1,2,i+1)
  sns.countplot(y=df[col], order = df[col].value_counts().index[:10]) # Top 10 Category
  plt.title(f"count plot of {col}")
  plt.tight_layout()
plt.show()
```




```
# Correlation Heatmap
plt.figure(figsize=(12,8))
corr_mat= df[num_colms].corr()
sns.heatmap(corr_mat, annot=True, fmt=".2f" ,cmap='coolwarm', linewidths=0.5)
plt.title('Correlation Heatmap')
plt.show()
```




```
 # Research Questions & Key Findings
 # 1. Which Brands Need Promotional or Pricing Adjustments?

# Step 1: Group by Brand and calculate metrics
brand_summary = df.groupby('Description').agg({
    'TotalSalesDollars': 'sum',
    'ProfitMargin': 'mean'
}).reset_index()

# Step 2: Define thresholds
low_Sales_threshold = brand_summary['TotalSalesDollars'].quantile(0.15)
high_margin_threshold = brand_summary['ProfitMargin'].quantile(0.85)
```

```
low_Sales_threshold
```

```
np.float64(560.299)
```

```
high_margin_threshold
```

```
np.float64(64.97017552750111)
```

```
# Step 3: Identify items needing promotion (high margin but low sales)
brands_to_promote = brand_summary[
    (brand_summary['TotalSalesDollars'] <= low_Sales_threshold) &
    (brand_summary['ProfitMargin'] >= high_margin_threshold)
]

# Results
brands_to_promote.sort_values(by='TotalSalesDollars')
```

	Description	TotalSalesDollars	ProfitMargin
6199	Santa Rita Organic Svgn Bl	9.99	66.466466
2369	Debauchery Pnt Nr	11.58	65.975820
2070	Concannon Glen Ellen Wh Zin	15.95	83.448276
2188	Crown Royal Apple	27.86	89.806174
6237	Sauza Sprklg Wild Berry Marg	27.96	82.153076
...	...	...	...
5074	Nanbu Bijin Southern Beauty	535.68	76.747312
2271	Dad's Hat Rye Whiskey	538.89	81.851584
57	A Bichot Clos Marechaudes	539.94	67.740860
6245	Sbragia Home Ranch Merlot	549.75	66.444748
3326	Goulee Cos d'Estournel 10	558.87	69.434752

198 rows × 3 columns

```
# 2. Who Are The Top Vendors And Brands By Highest Sales?

# Top Vendors by Total Sales
top_vendors = df.groupby('VendorName')['TotalSalesDollars'].sum().nlargest(10)

# Top Brands (Descriptions) by Total Sales
top_brands = df.groupby('Description')['TotalSalesDollars'].sum().nlargest(10)
```

```
top_vendors
```

VendorName	TotalSalesDollars
DIAGEO NORTH AMERICA INC	6.799010e+07
MARTIGNETTI COMPANIES	3.933036e+07
PERNOD RICARD USA	3.206320e+07
JIM BEAM BRANDS COMPANY	3.142302e+07
BACARDI USA INC	2.485482e+07
CONSTELLATION BRANDS INC	2.421875e+07
E & J GALLO WINERY	1.839990e+07
BROWN-FORMAN CORP	1.824723e+07
ULTRA BEVERAGE COMPANY LLP	1.650254e+07
M S WALKER INC	1.470646e+07

dtype: float64

```
top_brands
```

Description	TotalSalesDollars
Jack Daniels No 7 Black	7964746.76
Tito's Handmade Vodka	7399657.58
Grey Goose Vodka	7209608.06
Capt Morgan Spiced Rum	6356320.62
Absolut 80 Proof	6244752.03
Jameson Irish Whiskey	5715759.69
Ketel One Vodka	5070083.56
Baileys Irish Cream	4150122.07
Kahlua	3604858.66
Tanqueray	3456697.90

dtype: float64

```
# to conver the values of top brands in millions or k to make it more readable

def format_sales(value):
    if value >= 1_000_000:
        return f"{value/1_000_000:.2f}M"
    elif value >= 1_000:
        return f"{value/1_000:.2f}K"
    else:
        return str(value)

top_brands.apply(lambda x: format_sales(x))
```

Description	TotalSalesDollars
Jack Daniels No 7 Black	7.96M
Tito's Handmade Vodka	7.40M
Grey Goose Vodka	7.21M
Capt Morgan Spiced Rum	6.36M
Absolut 80 Proof	6.24M
Jameson Irish Whiskey	5.72M
Ketel One Vodka	5.07M
Baileys Irish Cream	4.15M
Kahlua	3.60M
Tanqueray	3.46M

dtype: object

```
# 3. Who Are the Top Vendors by Purchase Contribution?

vendor_performance = df.groupby('VendorName').agg({'TotalPurchaseDollars':'sum', 'GrossProfit':'sum', 'TotalSalesDollars':'sum'}).reset_inde


vendor_performance['PurchaseContribution%'] = (vendor_performance['TotalPurchaseDollars'] / vendor_performance['TotalPurchaseDollars'].sum()
vendor_performance = round(vendor_performance.sort_values('PurchaseContribution%', ascending=False),0)


# Display Top 10 Vendors
top_vendors = vendor_performance.head(10)
top_vendors['TotalSalesDollars'] = top_vendors['TotalSalesDollars'].apply(format_sales)
top_vendors['TotalPurchaseDollars'] = top_vendors['TotalPurchaseDollars'].apply(format_sales)
top_vendors['GrossProfit'] = top_vendors['GrossProfit'].apply(format_sales)
top_vendors
```

	VendorName	TotalPurchaseDollars	GrossProfit	TotalSalesDollars	PurchaseContribution%
25	DIAGEO NORTH AMERICA INC	50.10M	17.89M	67.99M	16.0
57	MARTIGNETTI COMPANIES	25.50M	13.83M	39.33M	8.0
68	PERNOD RICARD USA	23.85M	8.21M	32.06M	8.0
46	JIM BEAM BRANDS COMPANY	23.49M	7.93M	31.42M	8.0
6	BACARDI USA INC	17.43M	7.42M	24.85M	6.0
20	CONSTELLATION BRANDS INC	15.27M	8.95M	24.22M	5.0
11	BROWN-FORMAN CORP	13.24M	5.01M	18.25M	4.0
30	E & J GALLO WINERY	12.07M	6.33M	18.40M	4.0
106	ULTRA BEVERAGE COMPANY LLP	11.17M	5.34M	16.50M	4.0
53	M S WALKER INC	9.76M	4.94M	14.71M	3.0

Next steps: Generate code with top_vendors View recommended plots New interactive sheet

```
# Analyse the Top 10 vendor Total contribution in the Total Purchase

top_vendors['PurchaseContribution%'].sum()
```

```
np.float64(66.0)
```

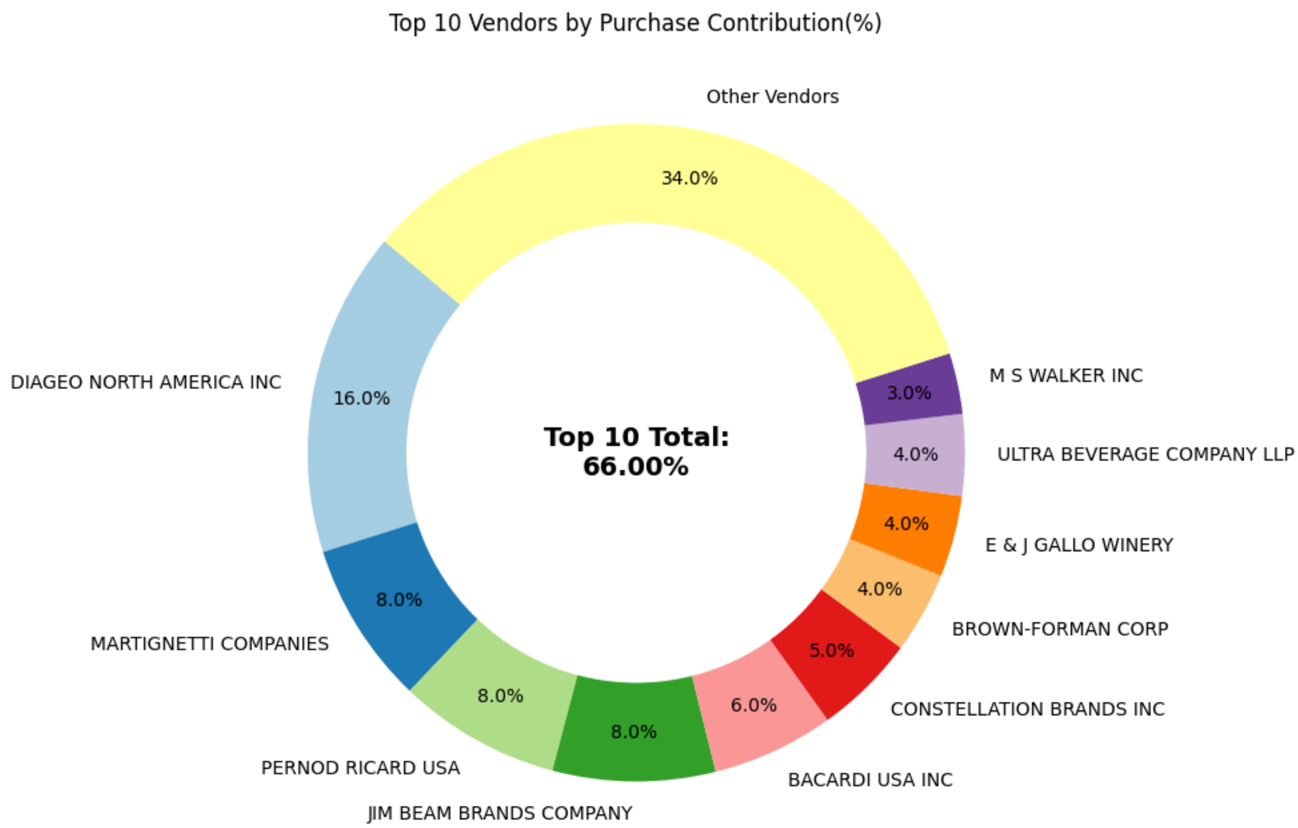
```
top_vendors['Cumulative_Contribution%'] = top_vendors['PurchaseContribution%'].cumsum()
top_vendors
```

	VendorName	TotalPurchaseDollars	GrossProfit	TotalSalesDollars	PurchaseContribution%	Cumulative_Contribution%
25	DIAGEO NORTH AMERICA INC	50.10M	17.89M	67.99M	16.0	16.0
57	MARTIGNETTI COMPANIES	25.50M	13.83M	39.33M	8.0	24.0
68	PERNOD RICARD USA	23.85M	8.21M	32.06M	8.0	32.0
46	JIM BEAM BRANDS COMPANY	23.49M	7.93M	31.42M	8.0	40.0
6	BACARDI USA INC	17.43M	7.42M	24.85M	6.0	46.0
20	CONSTELLATION BRANDS INC	15.27M	8.95M	24.22M	5.0	51.0
11	BROWN-FORMAN CORP	13.24M	5.01M	18.25M	4.0	55.0
30	E & J GALLO WINERY	12.07M	6.33M	18.40M	4.0	59.0

Next steps: Generate code with top_vendors View recommended plots New interactive sheet

```
vendors = list (top_vendors['VendorName'].values)
purchase_contribution = list (top_vendors['PurchaseContribution%'].values)
total_contribution = sum(purchase_contribution)
remaining_contribution = 100-total_contribution

# append other vendor category

vendors.append('Other Vendors')
purchase_contribution.append(remaining_contribution)

# Donut Chart

fig, ax = plt.subplots(figsize =(8,8))
wedges, texts, autotexts= ax.pie(purchase_contribution,
                                labels = vendors,
                                autopct = "%1.1f%%",
                                 startangle=140, pctdistance= 0.85, colors = plt.cm.Paired.colors)

# Draw a White Circle in the center to create a donut effect

centre_circle = plt.Circle((0,0), 0.70, fc='white')
fig = plt.gcf()
```

```
fig.gca().add_artist(centre_circle)

# Add the Total Contribution annotation in the centre

plt.text(0,0, f"Top 10 Total:\n{total_contribution:.2f}%", fontsize = 14, fontweight = 'bold', ha = 'center', va = 'center')
plt.title('Top 10 Vendors by Purchase Contribution(%)')
plt.show()
```




```
# 4. Does Bulk Purchasing Lower Unit Costs?

df['UnitPurchasePrice'] = df['TotalPurchaseDollars'] / df['TotalPurchaseQuantity']
df['OrderSize'] = pd.qcut(df['TotalPurchaseQuantity'], q=3, labels=['Small', 'Medium', 'Large'])
df.groupby('OrderSize')['UnitPurchasePrice'].mean()
```

OrderSize	UnitPurchasePrice
Small	39.057543
Medium	15.486414
Large	10.777625

dtype: float64

```
# 5. Which Vendors Have Low Inventory Turnover?

df[df.StockTurnover <1].groupby('VendorName')[['StockTurnover']].mean().sort_values('StockTurnover').head(10)
```

VendorName	StockTurnover
ALISA CARR BEVERAGES	0.615385
HIGHLAND WINE MERCHANTS LLC	0.708333
PARK STREET IMPORTS LLC	0.751306
Circa Wines	0.755676
Dunn Wine Brokers	0.766022
CENTEUR IMPORTS LLC	0.773953
SMOKY QUARTZ DISTILLERY LLC	0.783835
TAMWORTH DISTILLING	0.797078
THE IMPORTED GRAPE LLC	0.807569
WALPOLE MTN VIEW WINERY	0.820548

```
# 6. Do Profit Margins Differ by Vendor Type?
Top_threshold = df["TotalSalesDollars"].quantile(0.75)
low_threshold = df["TotalSalesDollars"].quantile(0.25)


top_vendors = df[df["TotalSalesDollars"] >= Top_threshold]['ProfitMargin'].dropna()
low_vendors = df[df["TotalSalesDollars"] <= low_threshold]['ProfitMargin'].dropna()


top_vendors
```

	ProfitMargin
0	25.297693
1	21.062810
2	24.675786
3	27.139908
4	28.412764
...	...
3966	79.684817
4170	85.782102
5530	93.085860
5773	95.012530
5945	94.271857

2142 rows × 1 columns

dtype: float64

```
low_vendors
```

	ProfitMargin
6728	4.111764
6761	6.145626
6825	12.007271
6828	1.677308
6861	7.239599
...	...
10687	83.448276
10688	96.436186
10689	25.252525
10690	98.974037
10691	99.166079

2142 rows × 1 columns

dtype: float64

```
def conf_intv(data,confidence = 0.95):

  mean_value = np.mean(data)
  std_err = np.std(data, ddof=1) / np.sqrt(len(data))
  t_critical = stats.t.ppf((1 + confidence) / 2, df=len(data) - 1)
  margin_of_error = t_critical * std_err
  confidence_interval = (mean_value, mean_value - margin_of_error, mean_value + margin_of_error)
  return confidence_interval


top_mean, top_lower, top_upper = conf_intv(top_vendors)
low_mean, low_lower, low_upper = conf_intv(low_vendors)

print(f"Top Vendors 95% CI: ({top_lower:.2f}, {top_upper:.2f}), Mean: {top_mean:.2f}")
print(f"Low Vendors 95% CI: ({low_lower:.2f}, {low_upper:.2f})), Mean: {low_mean:.2f}")
```

```
Top Vendors 95% CI: (30.74, 31.61), Mean: 31.17
Low Vendors 95% CI: (40.48, 42.62)), Mean: 41.55
```

```
 # 7. Are the Margin Differences Statistically Significant?


print(f"Top Vendors 95% CI: ({top_lower:.2f}, {top_upper:.2f}), Mean: {top_mean:.2f}")
print(f"Low Vendors 95% CI: ({low_lower:.2f}, {low_upper:.2f})), Mean: {low_mean:.2f}")

plt.figure(figsize = (12,6))

# TOP Vendor Plot

sns.histplot(top_vendors, kde=True, color='blue', bins = 30, label = 'Top Vendors')
plt.axvline(top_lower, color='blue', linestyle='dashed', label=f'Top Lower: {top_lower: .2f}')
plt.axvline(top_upper, color='blue', linestyle='dashed', label=f'Top Upper: {top_upper: .2f}')
plt.axvline(top_mean, color='blue', linestyle='dashed', label=f'Top Mean: {top_mean:.2f}')

# Low Vendor Plot

sns.histplot(low_vendors, kde=True, color='red', bins = 30, label = 'Low Vendors')
plt.axvline(low_lower, color='red', linestyle='dashed', label=f'Low Lower: {low_lower: .2f}')
plt.axvline(low_upper, color='red', linestyle='dashed', label=f'Low Upper: {low_upper: .2f}')
plt.axvline(low_mean, color='red', linestyle='dashed', label=f'Low Mean: {low_mean:.2f}')

# Final Plot

plt.title ("Confidence Intervsl Comparision: Top Vendors vs Low Vendors (Profit Margin)")
plt.xlabel("Profit Margin (%)")
plt.ylabel("Frequency")
plt.legend()
plt.grid(True)
plt.show()
```

```
Top Vendors 95% CI: (30.74, 31.61), Mean: 31.17
Low Vendors 95% CI: (40.48, 42.62)), Mean: 41.55
```